1 Model Extensions

The main model of **??** assumes that all agents share the same set of parameters. In this appendix, we extend the model by relaxing some of its assumptions and assess sensitivity to modeling choices. The extensions and sensitivity analyses are:

- **Collision avoidance**: an added repulsive term that promotes separation between nearby agents.
- **Salience-weighted beacons**: beacon selection based on salience as well as proximity.
- **Time-varying influence**: allow the balance between external and internal influence to change within a trial.
- **Individual heterogeneity**: allow agents to have different influence weights and sensing radii drawn from shared population distributions.
- **Sensing radius identification**: compare parameter recovery when the shared sensing radius is inferred or fixed.
- **Prior sensitivity**: assess parameter recovery and calibration under alternative prior distributions.

For each extension, we followed the simulation-based inference workflow of **??**. The collision, salience, and time-varying experiments infer a shared radius $r \sim \text{LogNormal}(0, 0.5)$, drawn once per trial. Their inputs comprise positions, headings, angular velocities, and neighbor counts at fixed reference distances $\{0.5, 1, 2, 4\}$ room units; no input statistics are evaluated at the unknown true radius. These experiments use relative-heading steering and additive diffusive heading noise, with η the noise amplitude. Trials contain 49 agents observed for 60 seconds at $\Delta t = 0.1$ seconds.

All recovery analyses are evaluated on 300 held-out simulations. Online analyses, including the individual-heterogeneity, fixed-radius, and prior-sensitivity comparisons, use 300 epochs of 50 batches of 32 fresh simulations (480,000 simulations), one training seed, and 1,000 posterior draws per test simulation. The collision, salience, and scalar time-varying models use a transformer summary network and flow-matching posterior estimator. The complementary Superstats trajectory analysis instead uses 30-second trials, offline training on 4,000 simulations, and 250 posterior draws per test simulation. Small differences between models should be interpreted descriptively.

We report normalized root-mean-square error (NRMSE), posterior contraction, and expected calibration error (ECE). Here, NRMSE is normalized by prior-only prediction error, rather than the range normalization in the original simulation study; NRMSE and contraction use the diagnostic implementation's median aggregation across test simulations. The individual-heterogeneity analysis uses a separate observation design and reports raw RMSE of posterior medians, empirical 95% interval coverage, and mean variance contraction. Its scores are therefore reported separately.

1.1 Collision Avoidance

As described in **??**, we used the Vicsek model as our base formulation of internal influence in **??**. This interaction aligns agents with their neighbors but does not enforce separation. When collision avoidance is of interest, we add a repulsive term to the orientation update.

Let $\mathcal{N}_{a,t}^{\rho} = \{n \neq a : 0 < \|\mathbf{x}_{a,t} - \mathbf{x}_{n,t}\| \leq \rho\}$ collect neighbors within a personal-space radius ρ. This radius is specified separately from the sensing radius r. The separation vector sums unit vectors directed away from nearby neighbors, weighted by their proximity,

$$\mathbf{s}_{a,t} = \sum_{n \in \mathcal{N}_{a,t}^{\rho}} q_{an,t} \frac{\mathbf{x}_{a,t} - \mathbf{x}_{n,t}}{\|\mathbf{x}_{a,t} - \mathbf{x}_{n,t}\|}, \qquad q_{an,t} = 1 - \frac{\|\mathbf{x}_{a,t} - \mathbf{x}_{n,t}\|}{\rho}. \tag{1}$$

The weight vanishes at the personal-space boundary. For $\|\mathbf{s}_{a,t}\| > 0$, let $\psi_{a,t}$ be its bearing. The orientation update acquires the additional separation drift

$$\Delta\theta_{a,t}^{\text{sep}} = \kappa\|\mathbf{s}_{a,t}\| \operatorname{wrap}(\psi_{a,t} - \theta_{a,t})\,\Delta t, \tag{2}$$

where wrap returns the corresponding angular difference in $[-\pi, \pi)$. This contribution is zero when the separation vector is zero and is added before the simulator's turn-rate cap. It sits outside the modulation by w, allowing separation to counteract both beacon attraction and neighbor alignment. We fix $\rho = 0.4$ and $\kappa = 0.6$ in the experiment below.

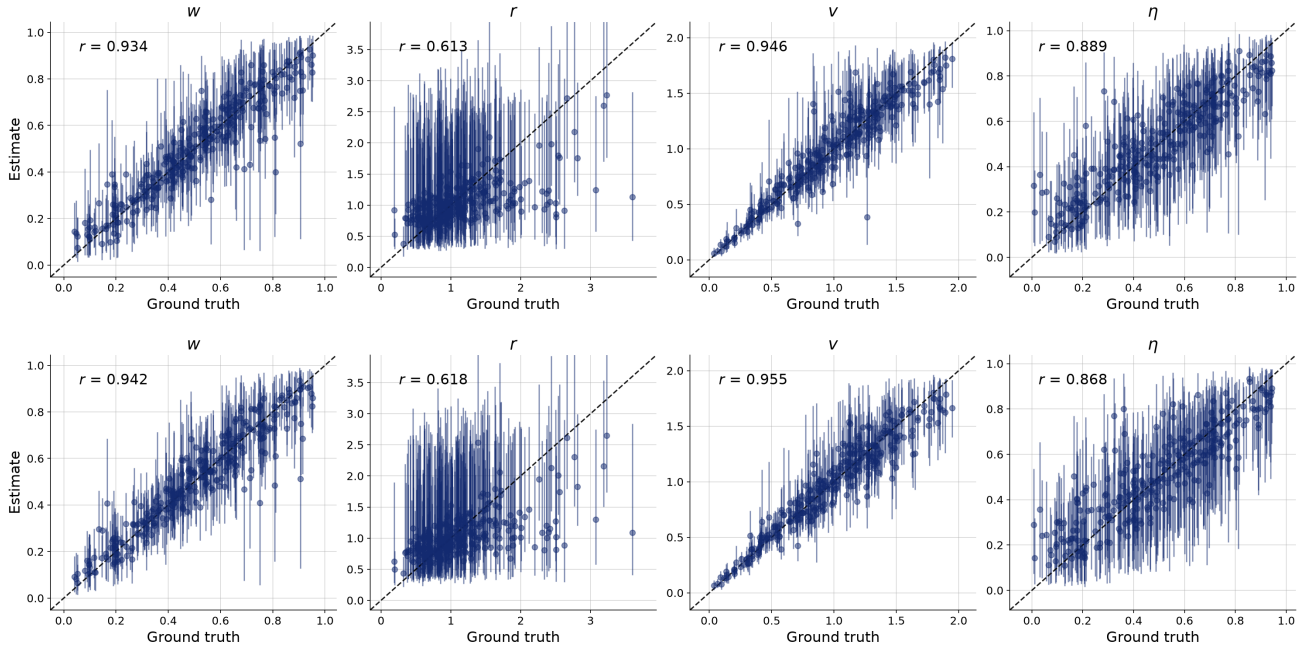


Figure 1: Parameter recovery for the variable-radius reference (*top*) and the collision-avoidance extension (*bottom*), with $\rho = 0.4$ and $\kappa = 0.6$.

Table 1: Parameter recovery and calibration for the variable-radius reference and the collision-avoidance extension with $\rho = 0.4$ and $\kappa = 0.6$.

		w	r	v	η
Reference	NRMSE	0.307	0.827	0.258	0.423
	Posterior contraction	0.910	0.302	0.936	0.826
	ECE	0.032	0.009	0.016	0.016
Collision avoidance	NRMSE	0.282	0.839	0.265	0.459
	Posterior contraction	0.923	0.337	0.942	0.777
	ECE	0.020	0.010	0.017	0.009

Simulation Study We compare the extension with the variable-radius reference without repulsion. Parameter recovery and empirical coverage are shown in **??** and **??**, with metrics in **??**. Influence-weight NRMSE decreases from 0.307 to 0.282. Speed recovery changes little, whereas noise recovery is weaker: η NRMSE increases from 0.423 to 0.459 and contraction falls from 0.826 to 0.777. Thus, this experiment does not show a general improvement in recovery from adding repulsion.

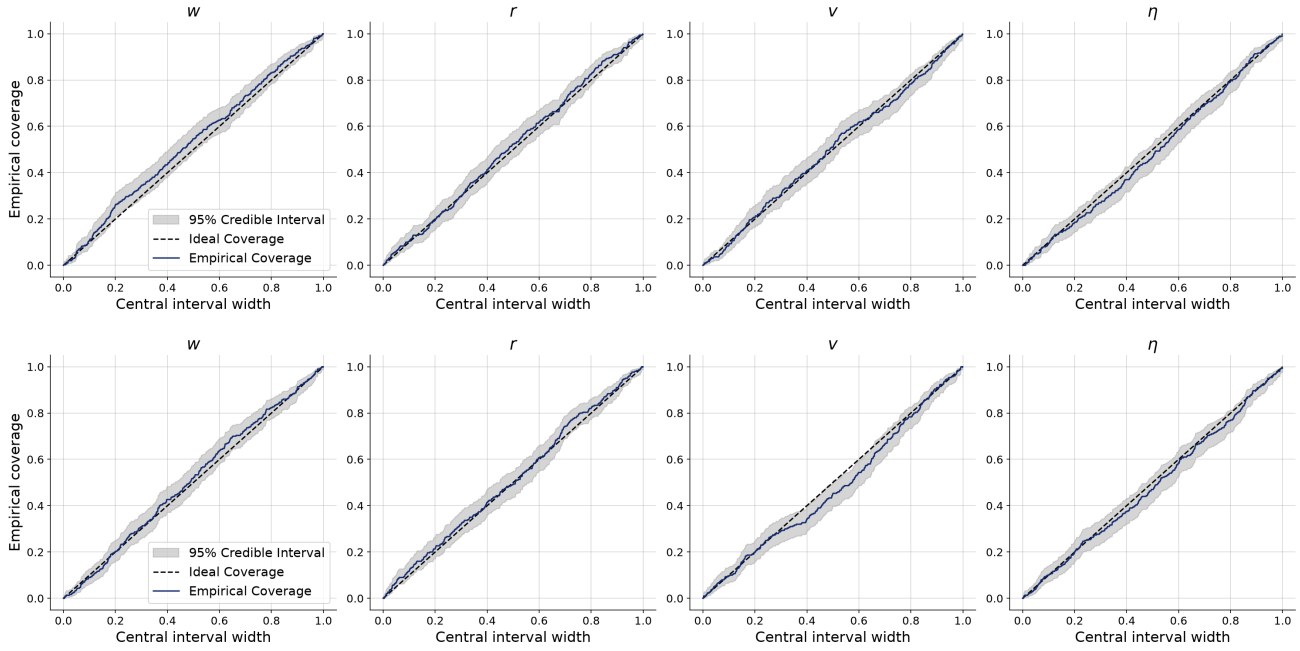


Figure 2: Empirical coverage for the variable-radius reference (*top*) and the collision-avoidance extension (*bottom*), with $\rho = 0.4$ and $\kappa = 0.6$.

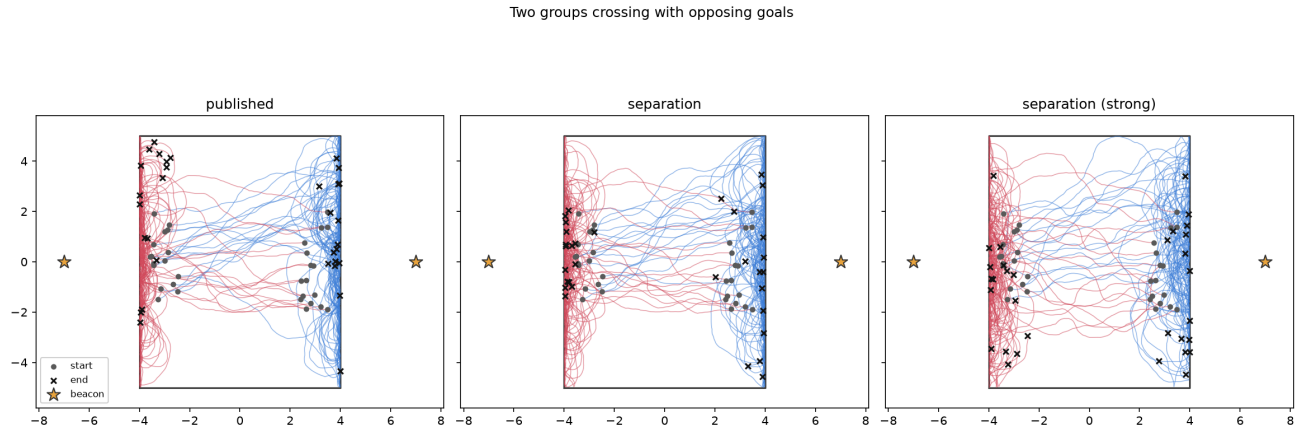


Figure 3: An illustrative crossing scenario with two groups of 16 agents assigned opposing beacon targets. Left: no repulsion; center: $\rho = 0.4$, $\kappa = 0.6$; right: $\rho = 0.6$, $\kappa = 1.5$. All panels use $w = 0.75$, $r = 2$, $v = 1$, $\eta = 0.5$, and a 30-second horizon. Colors distinguish groups, circles mark starting positions, and crosses mark endpoints. This fixed-parameter behavioral example is separate from the prior-averaged recovery experiment.

?? illustrates the behavioral effect in a separate crossing scenario. At the evaluated separation setting, the fraction of unordered agent-pair time points with distance below 0.4 falls from 0.048 to 0.033; stronger repulsion lowers it to 0.018. Close encounters remain, so the mechanism reduces crowding in this example without guaranteeing collision-free motion.

Radius recovery remains limited, with NRMSE 0.839 and contraction 0.337. ECE ranges from 0.009 to 0.020 in the extension. Behavioral separation and parameter recovery therefore provide distinct assessments of the extension.

1.2 Beacon Salience

The external influence of **??** selects the nearest beacon, so the approach target is determined by proximity. In more realistic scenarios, beacons can differ in their strength in guiding movement. For example, a large, vivid landmark or a louder bird chirp may compete for attention with closer, less salient objects.

To account for beacon salience, we assign each beacon $b \in B$ a salience score $s_b > 0$ and replace the selection rule with

$$b_{a,t} = \arg\max_{b \in B} \frac{s_b^{\alpha}}{\|\mathbf{x}_{a,t} - \mathbf{x}_{b,t}\|}, \tag{3}$$

where $\alpha \sim \text{Gamma}(2, 1)$ governs how strongly salience overrides proximity. At $\alpha = 0$, the rule reduces to nearest-beacon selection; at $\alpha = 1$, the score is salience divided by distance. Larger values give greater weight to salience. We use four beacons with fixed salience scores $(1, 1, 1, 8)$ and infer α alongside (w, r, v, η).

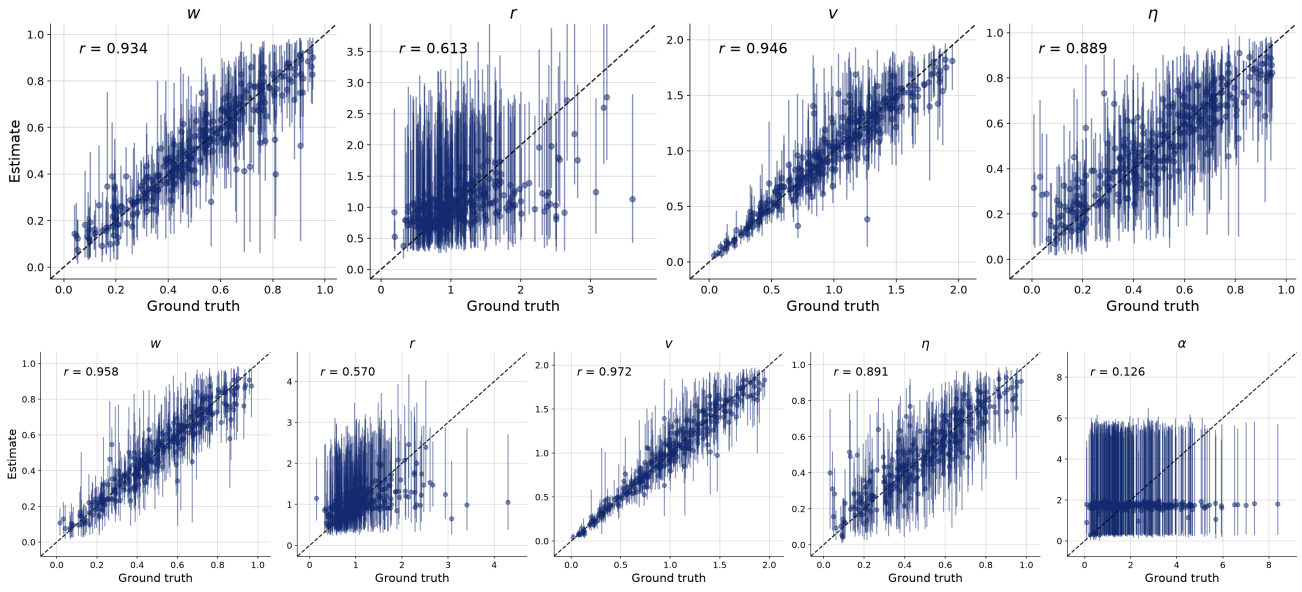


Figure 4: Parameter recovery for the variable-radius reference (*top*) and the beacon-salience extension (*bottom*).

Simulation Study We compare the extension with the variable-radius reference using **??**, **??**, and **??**. NRMSE is lower for each shared parameter in the salience configuration: w decreases from 0.307 to 0.264, r from 0.827 to 0.790, v from 0.258 to 0.205, and η from 0.423 to 0.404. These are descriptive differences between the two simulated configurations.

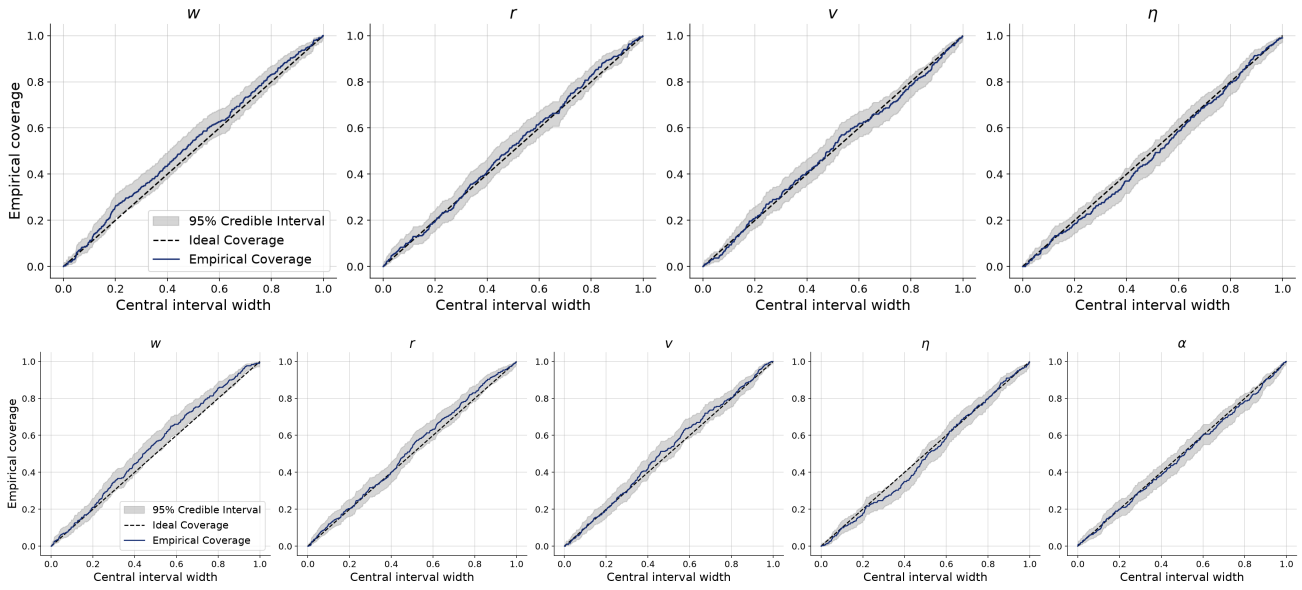


Figure 5: Empirical coverage for the variable-radius reference (*top*) and the beacon-salience extension (*bottom*).

Table 2: Parameter recovery and calibration for the variable-radius reference and the beacon-salience extension.

		w	r	v	η	α
Reference	NRMSE	0.307	0.827	0.258	0.423	—
	Posterior contraction	0.910	0.302	0.936	0.826	—
	ECE	0.032	0.009	0.016	0.016	—
Salience	NRMSE	0.264	0.790	0.205	0.404	0.957
	Posterior contraction	0.930	0.338	0.958	0.848	0.105
	ECE	0.041	0.021	0.011	0.011	0.010

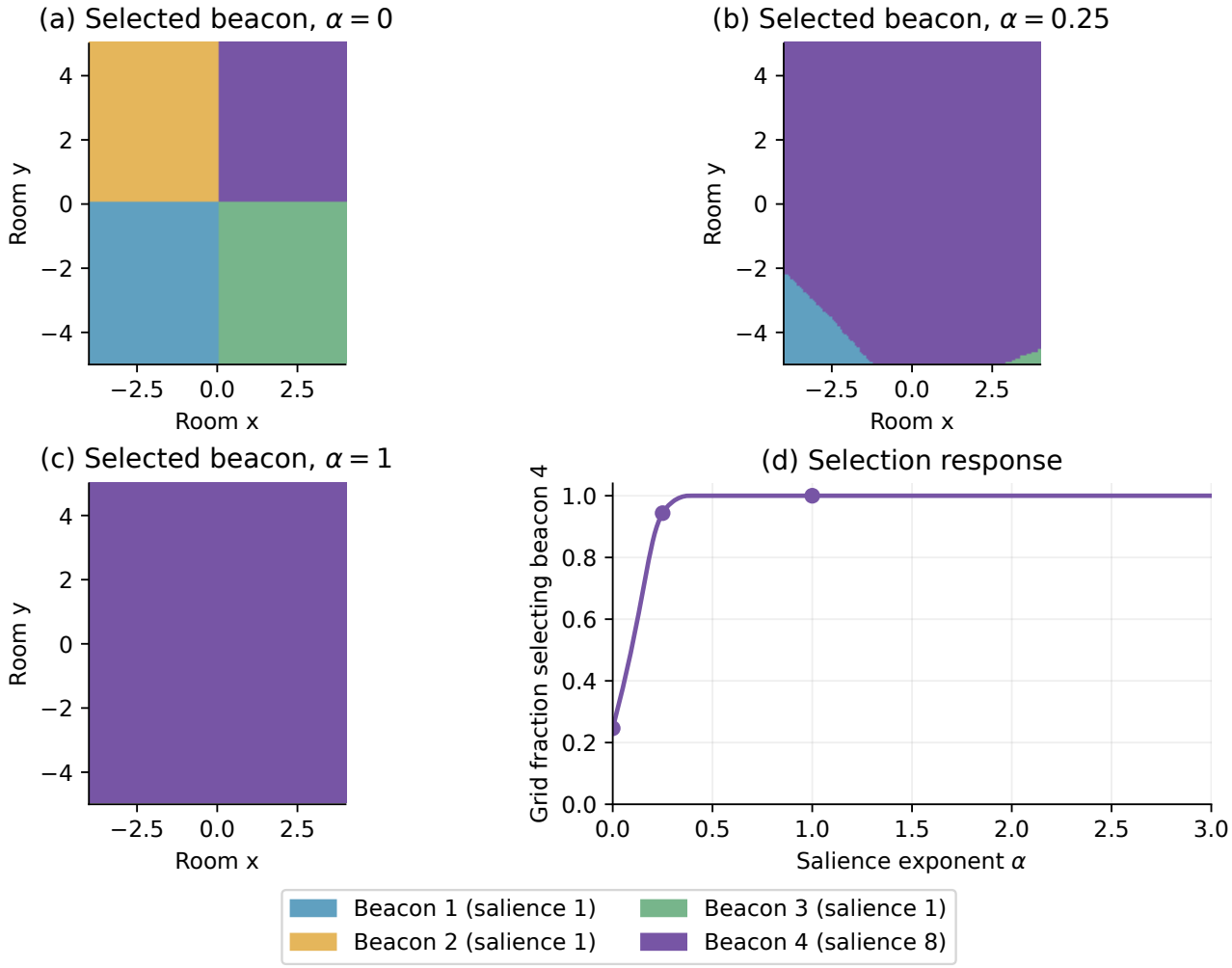


Figure 6: Illustration of the salience-selection rule for beacons at $(\pm 12, \pm 12)$ with strengths $(1, 1, 1, 8)$, the strongest at $(12, 12)$. Panels (a)–(c) show selected beacons over the 8-by-10 room for three exponents. Panel (d) shows the fraction of a fixed 121-by-151 room grid assigned to the strongest beacon as α increases. This deterministic layout illustrates the mechanism and is separate from the recovery test bank.

The salience exponent itself is weakly recovered, with NRMSE 0.957 and contraction 0.105, despite ECE 0.010. Thus, allowing salience to affect beacon selection does not imply that its exponent can be estimated precisely under this design.

?? shows how increasing α expands the strongest beacon's selection region. In this illustrative layout, the selection response saturates: further increases leave the selected beacon unchanged across the grid. This illustrates how distinct exponents can produce the same target choices, but does not establish structural non-identifiability or demonstrate that a narrower prior would resolve recovery.

1.3 Time-Varying Influence

The main model assumes that the influence weight remains constant throughout a trial. However, the balance between beacon-driven and neighbor-driven movement may change as agents move through the environment. To accommodate this possibility, we replace the constant weight w with a time-varying weight w_t.

Let $z_t = \mathrm{logit}(w_t)$. We model changes in influence using a random walk on the logit scale,

$$z_{t+\Delta t} = z_t + \tau\sqrt{\Delta t}\,\epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0,1), \qquad w_t = \frac{1}{1+\exp(-z_t)}, \tag{4}$$

with $z_0 = \mathrm{logit}(w_0)$, $w_0 \sim \mathrm{Beta}(2,2)$, and $\tau \sim 0.4\,\mathrm{Beta}(1,3)$. Here, w_0 is the initial influence weight and τ controls the magnitude of its fluctuations. The inverse-logit transformation keeps the weight between zero and one. Setting $\tau = 0$ recovers a constant influence weight.

This formulation allows influence to fluctuate without prescribing a direction of change. In particular, it does not impose a transition from exploration to following. The extension varies the shared influence weight over time; individual differences are considered separately in **??**.

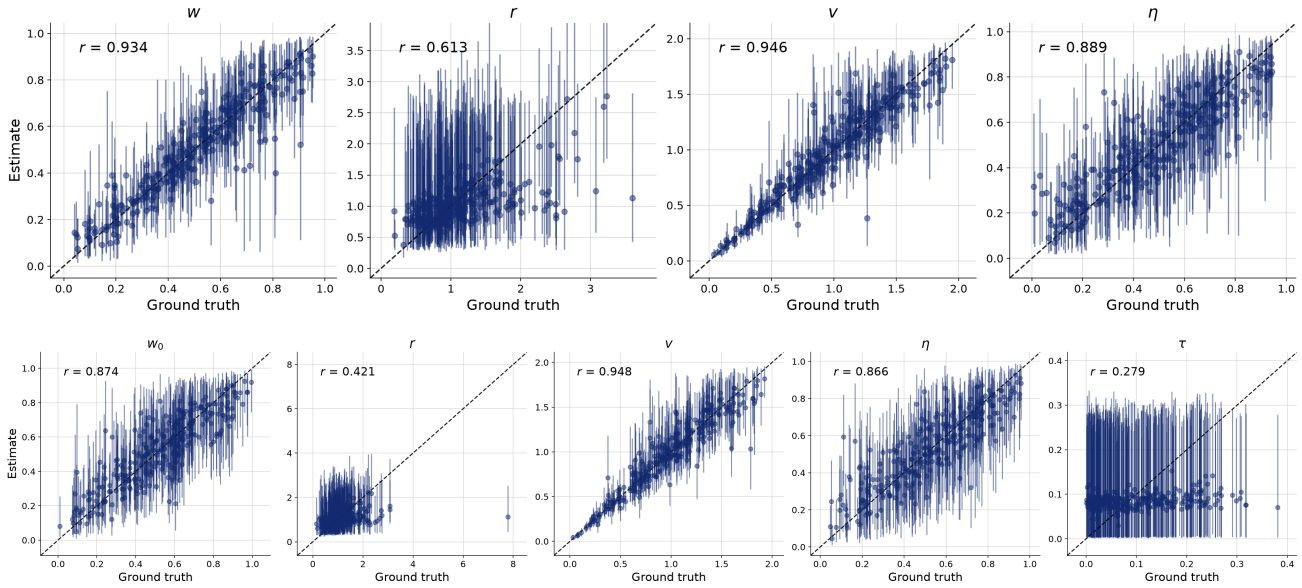


Figure 7: Parameter recovery for the variable-radius reference (*top*) and the time-varying influence extension (*bottom*). The reference estimates a constant w, whereas the extension estimates the initial weight w_0 and volatility τ alongside the shared radius, speed, and heading noise.

Simulation Study We infer the shared sensing radius alongside the time-varying influence parameters, with targets (w_0, r, v, η, τ). **??** and **??** compare recovery with the variable-radius reference.

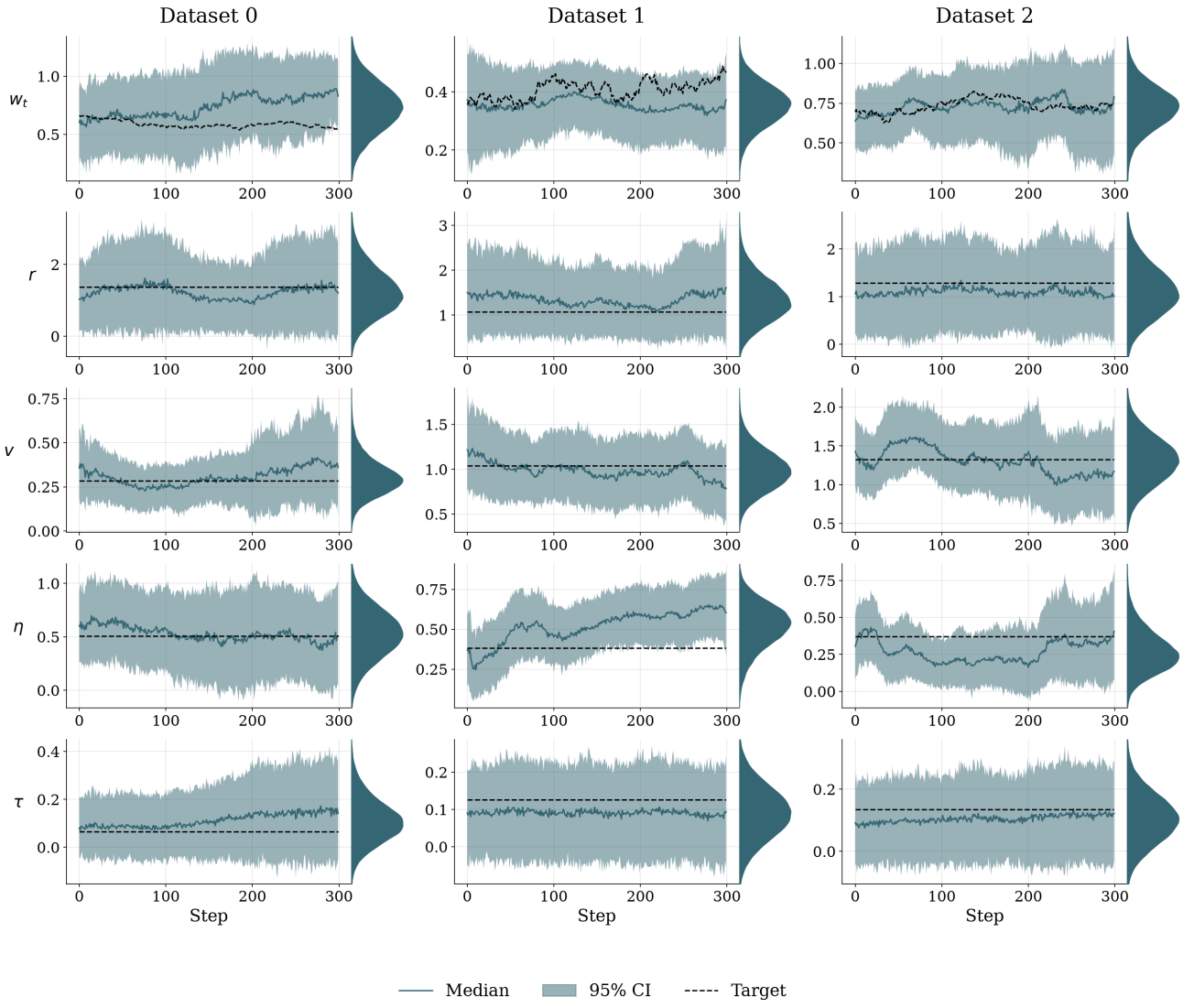


Figure 8: Time-resolved posterior estimates for the first three held-out datasets in the variable-radius Superstats analysis, selected by index. Solid lines show posterior medians, shaded bands pointwise 95% credible intervals, and dashed lines simulated targets. Only w_t varies in the generator; r, v, η, and τ are constant targets predicted at each step. The horizontal axis is simulation step, with $\Delta t = 0.1$ seconds.

Table 3: Parameter recovery and calibration for the time-varying influence extension and the variable-radius reference. Both infer r; the reference estimates a constant w, whereas the extension estimates w_0 and τ.

		w or w_0	r	v	η	τ
Reference	NRMSE	0.307	0.827	0.258	0.423	—
	Posterior contraction	0.910	0.302	0.936	0.826	—
	ECE	0.032	0.009	0.016	0.016	—
Time-varying	NRMSE	0.466	0.752	0.280	0.478	0.941
	Posterior contraction	0.797	0.445	0.914	0.767	0.090
	ECE	0.014	0.017	0.029	0.020	0.027

The initial weight has contraction 0.797, compared with 0.910 for the reference's constant weight. Volatility recovery is weaker, with NRMSE 0.941 and contraction 0.090. The experiment therefore provides less information about the magnitude of fluctuations than about the initial weight; this does not establish structural non-identifiability of τ.

?? shows posterior trajectories from the complementary Superstats analysis. Mean per-step contraction for w_t is 0.794, with mean per-step ECE 0.0315, but the examples also show tracking errors and broad intervals. The path can be informative while its volatility remains weakly recovered; these results concern simulated trajectories under a separate inference setup.

1.4 Individual Heterogeneity

The main model assigns the same influence weight and sensing radius to all agents. We relax this assumption by drawing individual weights and radii from shared population distributions,

$$\begin{aligned} \operatorname{logit}(w_a) &\sim \mathcal{N}\big(\operatorname{logit}(\mu_w), \sigma_w^2\big)\,, \\ \operatorname{logit}(r_a/R) &\sim \mathcal{N}\big(\operatorname{logit}(\mu_r/R), \sigma_r^2\big)\,, \qquad R = 10. \end{aligned} \tag{5}$$

The local draws are conditionally independent given the population parameters. Here, μ_w and μ_r are population medians on the physical scales, while σ_w and σ_r describe dispersion on their respective logit scales. The radius bound ensures $0 < r_a < R$. When both spreads are zero, all agents share the same weight and radius. The weight w_a multiplies beacon attraction; $1 - w_a$ multiplies neighbor alignment.

We use $\mu_w \sim \mathrm{Beta}(2,2)$, $\sigma_w \sim 1.5\,\mathrm{Beta}(1,3)$, $\mu_r \sim \mathrm{LogNormal}(\log 2.5, 0.35)$ truncated below R, and $\sigma_r \sim 0.75\,\mathrm{Beta}(2,2)$. Speed and heading noise remain shared, with $v \sim 2\,\mathrm{Beta}(2,2)$ and $\eta \sim \mathrm{Beta}(2,2)$. Each agent's weight and radius remain constant within a trial. Its own radius determines which neighbors contribute to its alignment, allowing asymmetric neighborhoods.

We jointly infer $(\mu_w, \sigma_w, \mu_r, \sigma_r, v, \eta)$ and all individual (w_a, r_a). Individual posteriors therefore depend on observed movement and the shared population model, yielding partial pooling. Unlike the preceding extensions, this controlled example uses 12 agents, 20-second trials, and four known beacons at $(-3,-4)$, $(-3,4)$, $(3,-4)$, and $(3,4)$ in an 8-by-10 room. Inputs are joint turning and motion summaries evaluated at fixed reference radii from 0.5 to 12 room units; no latent radii or weights enter the inputs. A flow-matching estimator conditions the joint posterior on these summaries.

Table 4: Recovery in the 12-agent partial-pooling model with individual weights and bounded sensing radii. RMSE uses posterior medians; the prior baseline predicts each parameter by its empirical prior median. Coverage is for marginal 95% intervals. Individual rows report arithmetic means of the 12 agent-specific metrics; contraction is the mean reduction in posterior variance relative to the prior-reference bank.

Parameter	RMSE	Prior-median RMSE	95% coverage (%)	Contraction
μ_w	0.035	0.227	98.7	0.953
σ_w	0.188	0.296	95.7	0.408
μ_r	0.488	1.021	93.7	0.736
σ_r	0.161	0.175	93.7	0.155
v	0.040	0.427	99.3	0.951
η	0.018	0.229	98.7	0.959
w_a (mean over agents)	0.054	0.240	95.8	0.928
r_a (mean over agents)	0.824	1.272	92.7	0.611

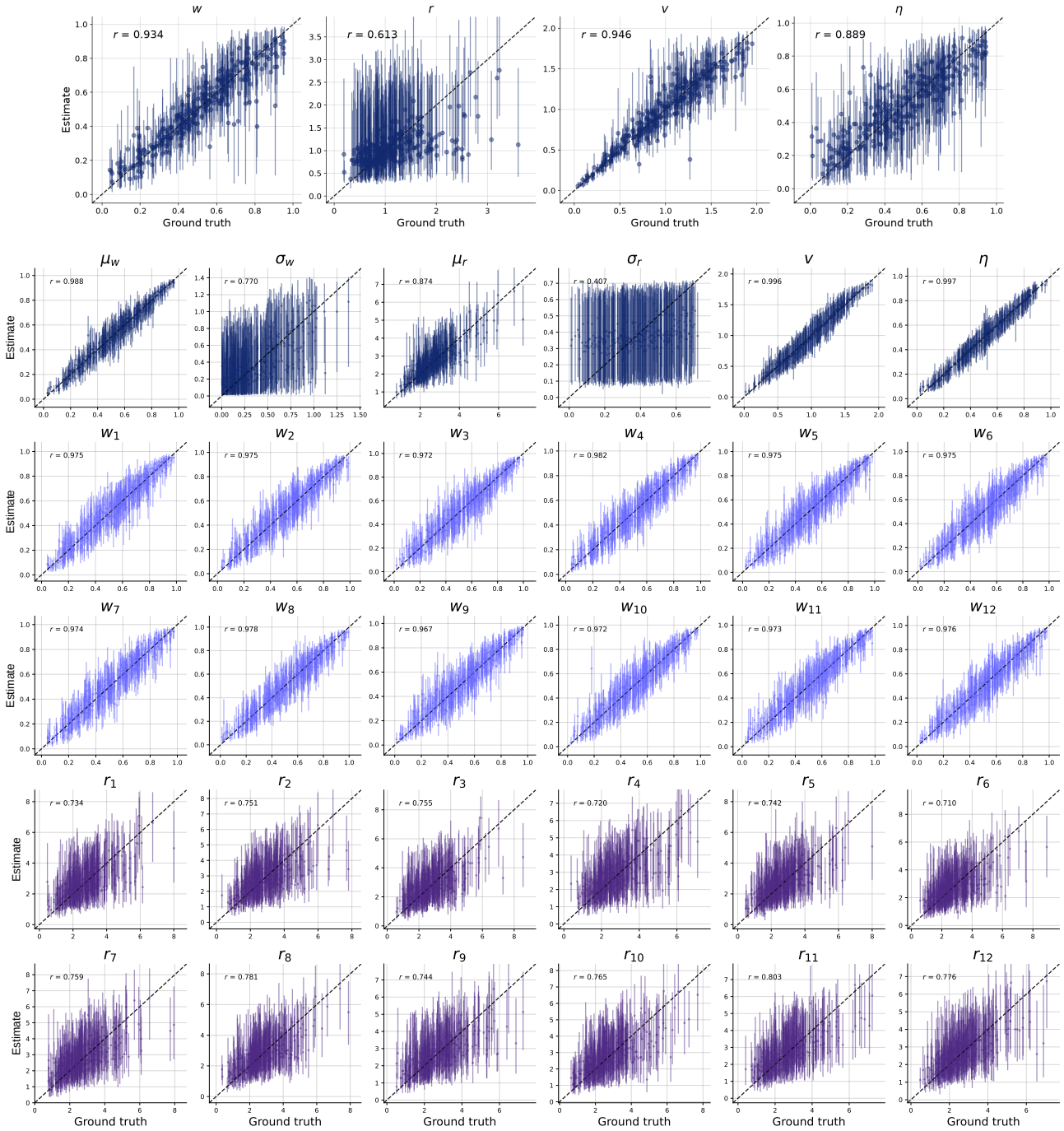


Figure 9: Parameter recovery for the variable-radius reference (*top*) and the 12-agent partial-pooling model (*bottom*). In the lower panel, the first row shows population and shared parameters, the next two rows individual weights, and the last two rows individual radii. The reference provides descriptive context: group size, priors, and observation features differ, so the panels do not constitute a matched comparison.

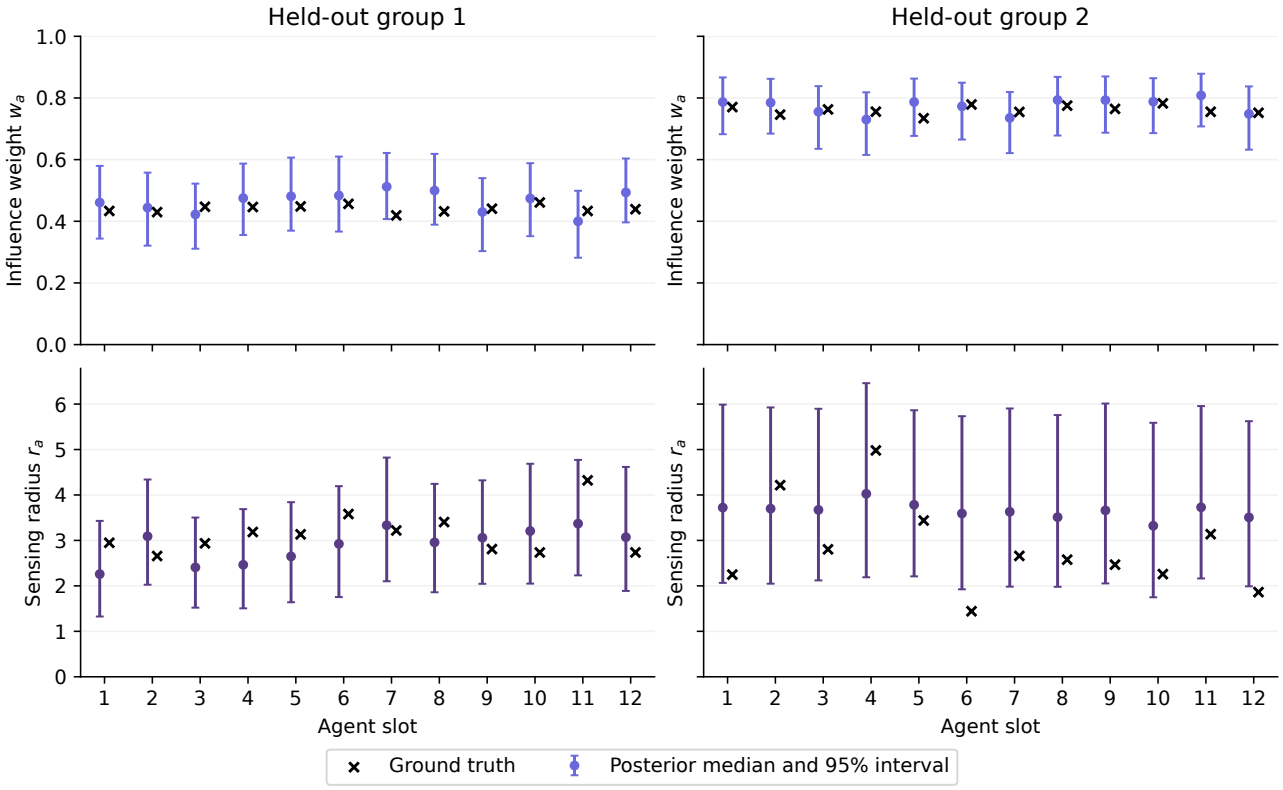


Figure 10: Individual posterior estimates for the first two held-out groups in the 12-agent hierarchy, selected by index. The upper row shows influence weights and the lower row sensing radii. Colored points and bars show posterior medians and marginal 95% credible intervals; black crosses show simulated truths. Agent slots identify individuals within each group, not persistent individuals across groups.

Simulation Study **??** shows overall recovery, while **??** displays individual estimates and uncertainty within two held-out groups. **??** reports the aggregate metrics. The reference comparison is descriptive because priors, features, and group size differ.

Individual-weight RMSE averages 0.054, compared with 0.240 for the prior-median predictor, with mean marginal 95% coverage of 95.8%. After centering weights within each group, RMSE is 0.049 against a zero-difference baseline of 0.088, correlation is 0.827, and 95% interval coverage is 96.4%. These results support recovery of individual weight differences beyond the group center.

Radius differences are less precisely recovered: centered RMSE is 0.685 against a zero-difference baseline of 0.755, with correlation 0.422 and 95% coverage of 91.4%. Population radius spread is also weakly constrained, with contraction 0.155. Thus, this hierarchy demonstrates partial pooling and informative weight differences, while individual-radius recovery and uncertainty remain limited. Agents within a group are dependent, so they are not independent test replications.

1.5 Sensing radius identification

Table 5: Parameter recovery and calibration with inferred or fixed sensing radius. The inferred-radius model uses $r \sim \text{LogNormal}(0, 0.5)$ and neighborhood summaries at fixed reference distances; fixed-radius models use summaries at their specified radius. Metrics evaluate each model under its own specification, so differences include the observation definitions and radius distribution as well as inference dimension. Dashes indicate that radius is not inferred.

		w	r	v	η
Inferred radius	NRMSE	0.307	0.827	0.258	0.423
	Posterior contraction	0.910	0.302	0.936	0.826
	ECE	0.032	0.009	0.016	0.016
Fixed $r = 2.5$	NRMSE	0.185	—	0.203	0.385
	Posterior contraction	0.968	—	0.957	0.847
	ECE	0.012	—	0.016	0.009
Fixed $r = 1.25$	NRMSE	0.208	—	0.168	0.335
	Posterior contraction	0.954	—	0.968	0.884
	ECE	0.009	—	0.042	0.011
Fixed $r = 4$	NRMSE	0.173	—	0.217	0.409
	Posterior contraction	0.970	—	0.954	0.820
	ECE	0.015	—	0.019	0.013

Simulation Study **??** shows limited recovery of the unknown shared radius, with NRMSE 0.827 and contraction 0.302, despite ECE 0.009. With $r = 2.5$ fixed, NRMSE decreases from 0.307 to 0.185 for w, from 0.258 to 0.203 for v, and from 0.423 to 0.385 for η, with greater contraction for all three parameters.

All three fixed-radius specifications yield lower NRMSE for the remaining parameters than the inferred-radius model, but no constant performs best across all diagnostics: $r = 1.25$ gives the lowest speed NRMSE and the highest speed ECE (0.042). Fixing radius is therefore a useful working specification whose performance remains conditional on the chosen value. Because observation definitions and test populations also differ, the gains cannot be attributed solely to removing radius from inference; these experiments do not establish structural non-identifiability of r.

1.6 Prior Sensitivity

We assess sensitivity to the priors for w, v, and η using the fixed-radius specification with $r = 2.5$. **??** lists the reference and five alternatives. The wide and concentrated configurations change all three priors together; the remaining configurations change only the speed or noise prior. The radius is a specified constant throughout this analysis; sensitivity to its value is reported in **??**.

Table 6: Prior specifications for the fixed-radius sensitivity analysis. All configurations fix $r = 2.5$ room units.

Configuration	w	v	η
Reference	$\text{Beta}(2, 2)$	$2\,\text{Beta}(2, 2)$	$\text{Beta}(2, 2)$
Wide	$\text{Beta}(1, 1)$	$2\,\text{Beta}(1, 1)$	$\text{Beta}(1, 1)$
Concentrated	$\text{Beta}(5, 5)$	$2\,\text{Beta}(5, 5)$	$\text{Beta}(5, 5)$
Slower speed	$\text{Beta}(2, 2)$	$0.05 + 0.95\,\text{Beta}(2, 2)$	$\text{Beta}(2, 2)$
Lower noise	$\text{Beta}(2, 2)$	$2\,\text{Beta}(2, 2)$	$\text{Beta}(2, 5)$
Higher noise	$\text{Beta}(2, 2)$	$2\,\text{Beta}(2, 2)$	$1.5\,\text{Beta}(2, 2)$

Table 7: Parameter recovery and calibration under alternative priors, conditional on $r = 2.5$. Each model is evaluated on simulations from its own prior specification. NRMSE is normalized by the corresponding prior-only prediction error; contraction is measured relative to that prior.

		w	v	η
Reference	NRMSE	0.185	0.203	0.385
	Posterior contraction	0.968	0.957	0.847
	ECE	0.012	0.016	0.009
Wide	NRMSE	0.190	0.146	0.358
	Posterior contraction	0.961	0.974	0.866
	ECE	0.014	0.026	0.006
Concentrated	NRMSE	0.201	0.253	0.577
	Posterior contraction	0.960	0.938	0.707
	ECE	0.013	0.014	0.009
Slower speed	NRMSE	0.224	0.207	0.422
	Posterior contraction	0.948	0.960	0.823
	ECE	0.016	0.010	0.015
Lower noise	NRMSE	0.215	0.217	0.583
	Posterior contraction	0.959	0.950	0.677
	ECE	0.008	0.023	0.015
Higher noise	NRMSE	0.198	0.183	0.274
	Posterior contraction	0.959	0.963	0.923
	ECE	0.028	0.021	0.023

Simulation Study **??** compares recovery under the reference, wide, and concentrated priors; **??** summarizes all configurations. Across the six specifications, ECE remains below 0.028, while NRMSE ranges from 0.185 to 0.224 for w, 0.146 to 0.253 for v, and 0.274 to 0.583 for η. Noise is the least accurately recovered parameter by this metric in every configuration.

Noise contraction varies from 0.677 under the lower-noise prior to 0.923 under the higher-noise prior, compared with 0.847 in the reference. These scores depend on both the prior variance and the

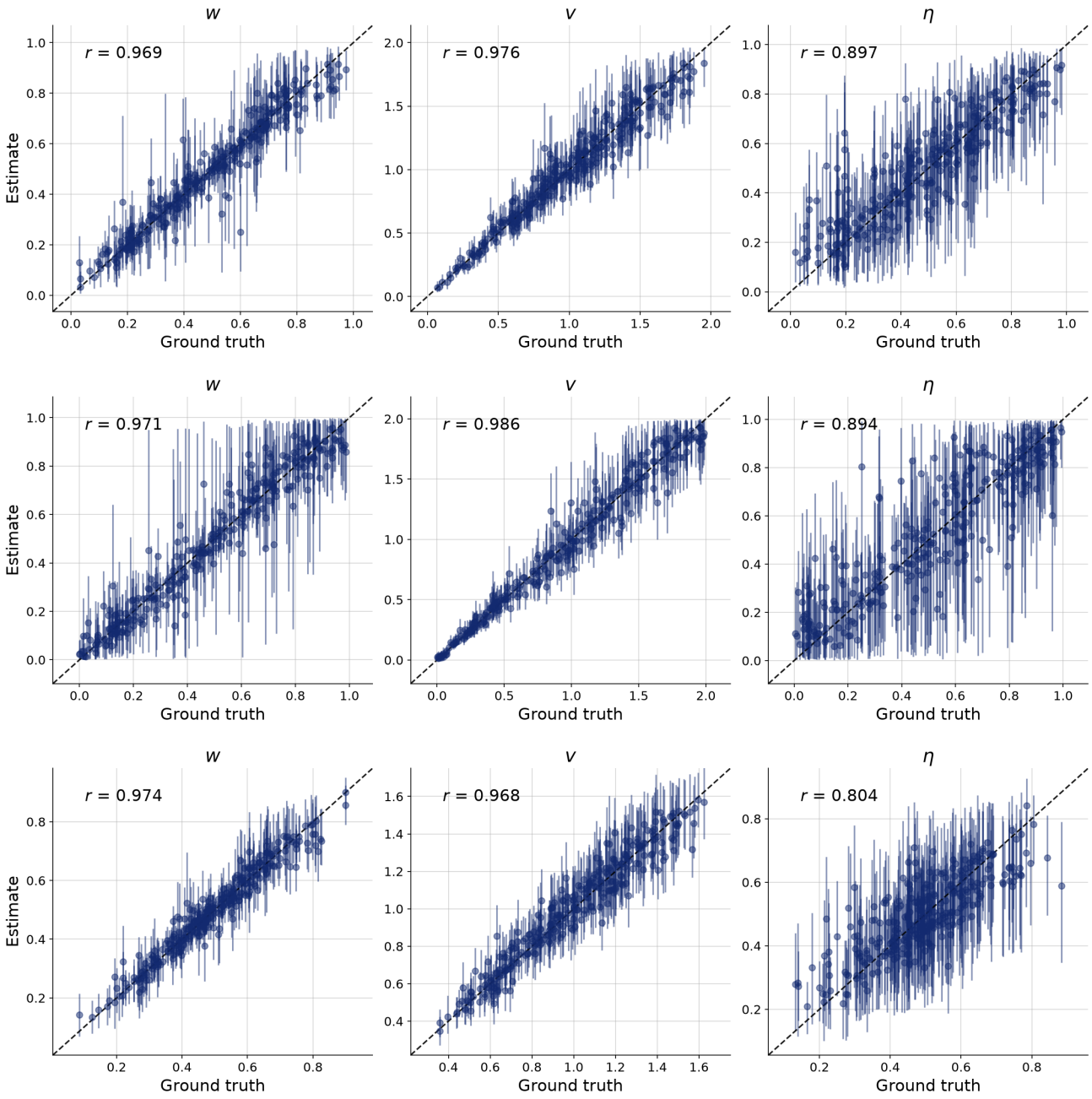


Figure 11: Parameter recovery under the fixed-radius reference priors (*top*), wide priors (*middle*), and concentrated priors (*bottom*). Each row shows w, v, and η. Axis ranges reflect the different prior populations; each estimator is evaluated under its own prior specification.

simulated parameter population; larger contraction alone does not establish greater absolute precision. Restricting speed gives NRMSE 0.207 versus 0.203 in the reference, despite lower ECE (0.010 versus 0.016).

These experiments support low marginal calibration error across the tested priors, while recovery varies with the specification. They assess inference over different prior-predictive populations and do not establish that posteriors for identical observations are insensitive to prior choice. The conclusions are conditional on the fixed radius and do not establish robustness to alternative priors on an unknown r.